

RNAseq Expression Specificity: SCZ GWAS Enrichment Results

This document summarizes the core RNA-seq-based cell-type enrichment results for schizophrenia (SCZ) genetic risk, and their intersection with case-control compositional changes from post-mortem snRNA-seq.

1. Cell-Type Specificity and GWAS Enrichment

Method

For each of 137 SEA-AD supertypes, cell-type specificity scores were computed (column-normalize to equalize library sizes, then row-normalize so each gene sums to 1.0 across types). MAGMA-style OLS regression tests whether genes specific to each type carry excess SCZ GWAS signal:

$$\text{GWAS_Z} \sim \text{specificity} + \log(\text{gene_size}) + \log(\text{n_snps}) + \log(N)$$

SEA-AD-Only Results (137 supertypes)

Metric	Value
Total supertypes tested	137
Genes tested	17,160
FDR-significant ($q < 0.05$)	44
Bonferroni-significant	17
GABAergic FDR-sig	41
Glutamatergic FDR-sig	3
Non-neuronal FDR-sig	0

Key finding: SCZ genetic risk is overwhelmingly concentrated in GABAergic interneurons (41/44 = 93% of significant types), with SST, Pvalb, Lamp5, Pax6, Vip, and Sncg subtypes all represented.

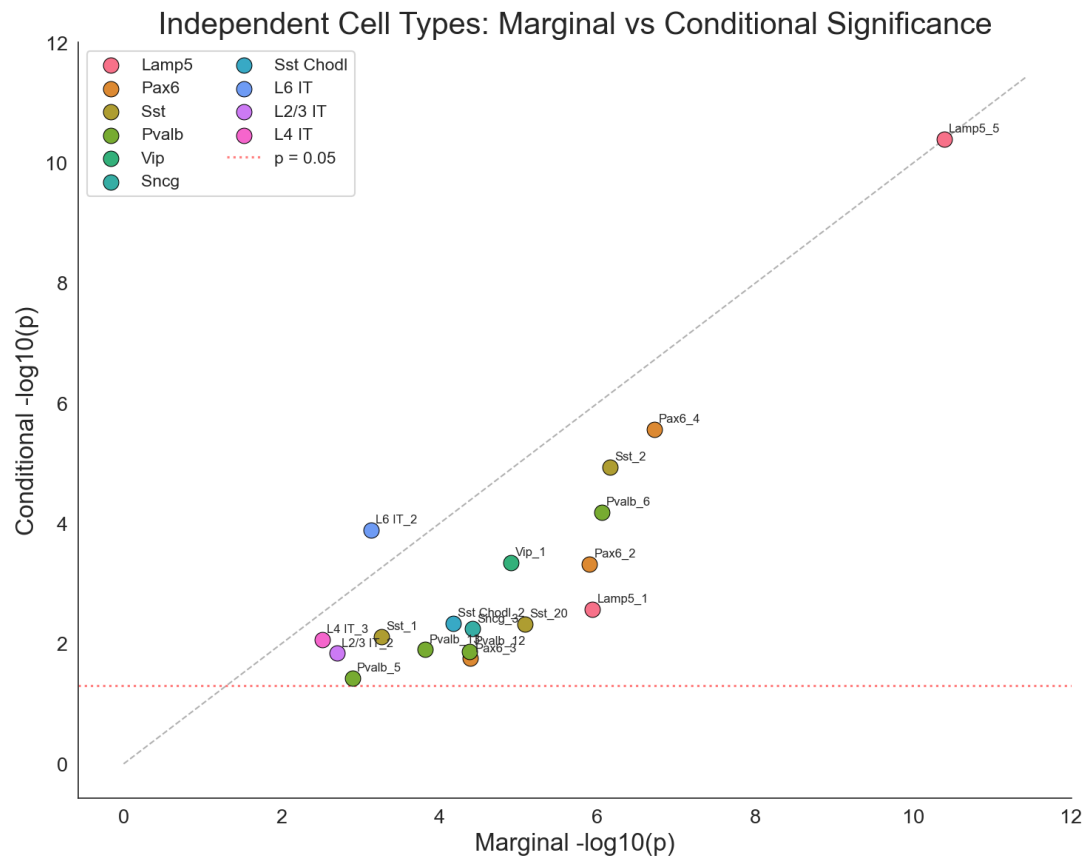


Figure 2. Conditional analysis. Each point is an independently enriched cell type; x-axis shows marginal significance, y-axis shows conditional significance after accounting for all previously accepted types. Points above the $p = 0.05$ line carry independent signal. Colored by subclass.

Step	Cell type	Subclass	Marginal p	Conditional p	Conditional beta
1	Lamp5_5	Lamp5	4.05e-11	4.05e-11	9.77
2	Pax6_4	Pax6	1.90e-07	2.69e-06	4.70
3	Sst_2	Sst	6.95e-07	1.16e-05	5.77
4	Pvalb_6	Pvalb	8.79e-07	6.49e-05	4.40
5	Lamp5_1	Lamp5	1.15e-06	2.65e-03	3.43
6	Pax6_2	Pax6	1.26e-06	4.77e-04	2.87
7	Sst_20	Sst	8.22e-06	4.68e-03	3.78
8	Vip_1	Vip	1.25e-05	4.45e-04	3.12
9	Sncg_3	Sncg	3.80e-05	5.66e-03	2.28
10	Pax6_3	Pax6	4.06e-05	1.77e-02	2.05
11	Pvalb_12	Pvalb	4.12e-05	1.34e-02	2.83
12	Sst Chodl_2	Sst Chodl	6.71e-05	4.57e-03	2.51
13	Pvalb_13	Pvalb	1.53e-04	1.26e-02	2.16
14	Sst_1	Sst	5.38e-04	7.64e-03	1.74
15	L6 IT_2	L6 IT	7.37e-04	1.28e-04	1.99

Step	Cell type	Subclass	Marginal p	Conditional p	Conditional beta
16	Pvalb_5	Pvalb	1.27e-03	3.71e-02	1.41
17	L2/3 IT_2	L2/3 IT	1.99e-03	1.44e-02	2.24
18	L4 IT_3	L4 IT	3.05e-03	8.56e-03	2.24

Interpretation: Even after conditioning, multiple subtypes within Sst (3 types), Pvalb (4 types), Pax6 (3 types), and Lamp5 (2 types) remain independently significant. This indicates genuine subtype-specific enrichment, not just shared interneuron gene programs. The three glutamatergic types (L6 IT_2, L2/3 IT_2, L4 IT_3) survive conditioning, suggesting a weaker but real excitatory neuron signal.

Gene Drivers: Example for Sst_2

For each enriched cell type, “driver genes” are identified as genes in the top 10% of cell-type specificity AND GWAS FDR < 0.05. These are the genes that make a cell type enriched for SCZ risk.

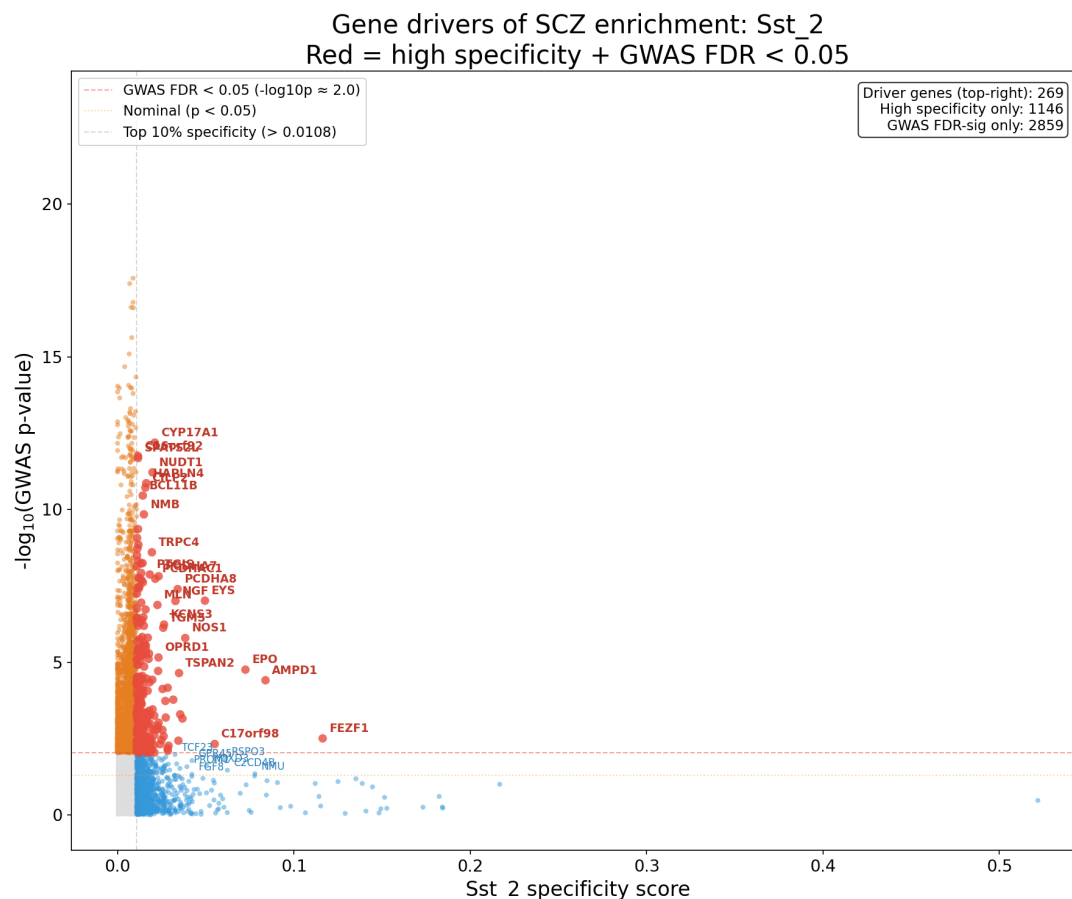


Figure 3. Gene driver scatter for Sst_2. X-axis: cell-type specificity score; Y-axis: $-\log_{10}(\text{GWAS p-value})$. Red points (top-right quadrant) are driver genes — both highly specific to Sst_2 and strongly associated with SCZ. 269 driver genes identified. Key genes labeled include CYP17A1, GRIN2D, MAD1L1, and CACNA1C.

2. RBH Combined Taxonomy (SEA-AD + Siletti)

Motivation

SEA-AD covers neocortical cell types with fine resolution. The Siletti et al. (2023) whole-brain atlas adds 461 clusters spanning hippocampus, amygdala, thalamus, and other regions. MetaNeighbor reciprocal best hits (RBH) identified 95 Siletti clusters that are transcriptomically equivalent to SEA-AD types. The combined taxonomy keeps all 137 SEA-AD types plus 366 novel Siletti clusters = **503 total types**.

RBH Combined Enrichment Results

Metric	Value
Total types	503
SEA-AD types	137
Novel Siletti clusters	366
FDR-significant	120 (68 SEA-AD + 52 Siletti)
Bonferroni-significant	54

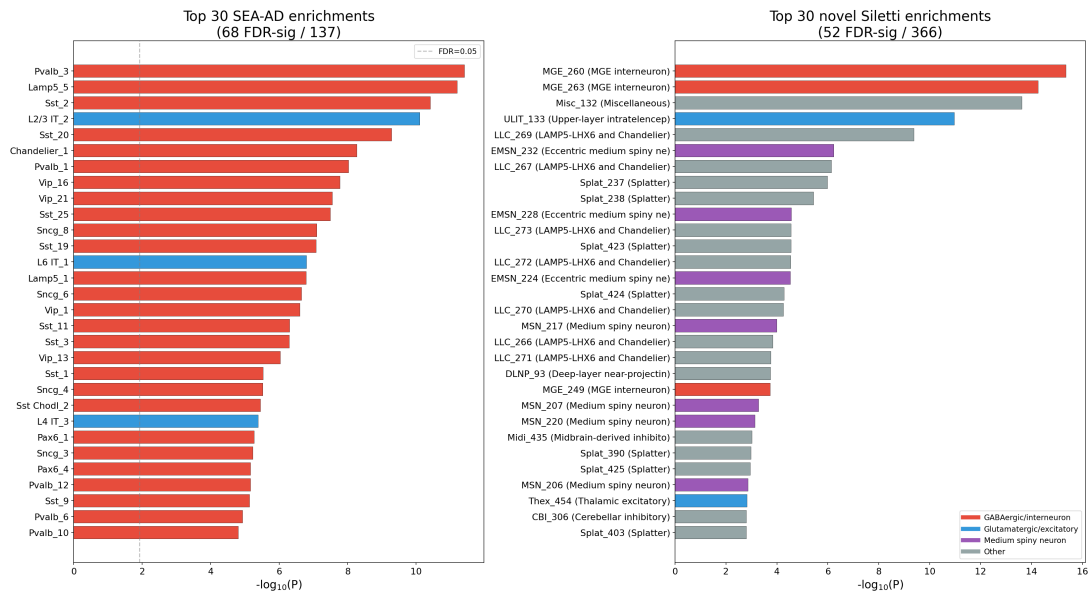


Figure 4. Top 30 enrichments from the RBH combined taxonomy, split by source. Left: SEA-AD types (68 FDR-sig) dominated by GABAergic interneurons. Right: novel Siletti clusters (52 FDR-sig) dominated by MGE interneuron clusters and LAMP5-LHX6 types from subcortical brain regions. Colored by broad cell class.

Top 15 Enriched Types (RBH Combined)

Rank	Cell type	Beta	FDR p-value	Source
1	MGE_260	61.31	2.23e-13	Siletti
2	MGE_263	54.62	1.40e-12	Siletti
3	Misc_132	38.17	3.98e-12	Siletti

Rank	Cell type	Beta	FDR p-value	Source
4	Pvalb_3	47.22	4.85e-10	SEA-AD
5	Lamp5_5	40.00	6.43e-10	SEA-AD
6	ULIT_133	47.68	9.19e-10	Siletti
7	Sst_2	38.35	2.75e-09	SEA-AD
8	L2/3 IT_2	34.07	4.98e-09	SEA-AD
9	LLC_269	43.79	2.33e-08	Siletti
10	Sst_20	39.16	2.64e-08	SEA-AD
11	Chandelier_1	30.82	2.49e-07	SEA-AD
12	Pvalb_1	29.93	4.01e-07	SEA-AD
13	Vip_16	30.85	6.52e-07	SEA-AD
14	Vip_21	24.82	1.00e-06	SEA-AD
15	Sst_25	33.71	1.09e-06	SEA-AD

Key finding: The top-ranked types include Siletti MGE (medial ganglionic eminence) clusters that represent interneuron progenitors or non-neocortical interneurons. These whole-brain interneuron populations carry even stronger SCZ signal than any individual neocortical subtype, consistent with SCZ being a brain-wide interneuron disorder rather than a purely cortical one.

3. SST Interneurons: Deep Dive

SST interneurons warrant focused analysis because they are the most enriched subclass and include multiple independently significant subtypes. The figure below uses enrichment results from the **RBH combined taxonomy** (503 types: 137 SEA-AD + 366 novel Siletti) rather than the SEA-AD-only analysis, providing a more stringent test of SST enrichment in a broader cell-type context. Cortical depth is from SEA-AD MERFISH spatial transcriptomics (weighted average of MERFISH + Xenium platforms; 0 = pial surface, 1 = white matter boundary).

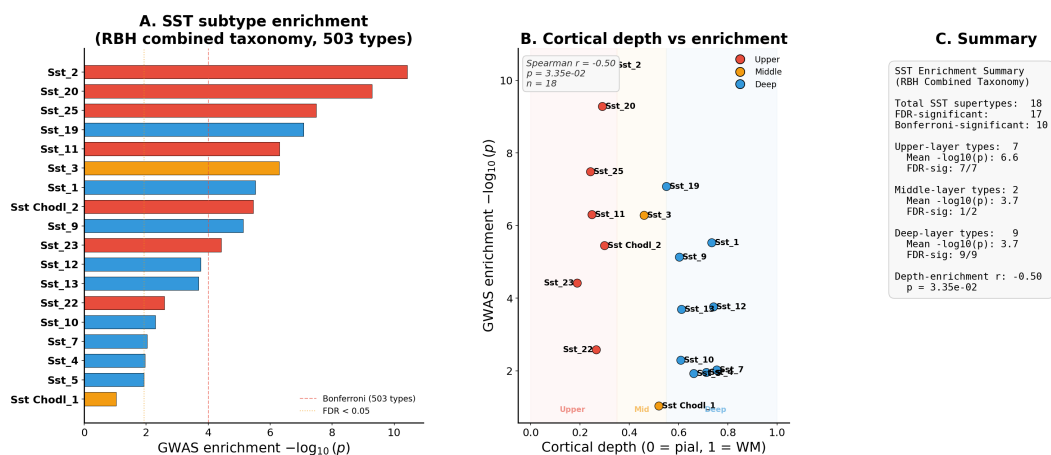


Figure 5. SST subtype deep dive using RBH combined taxonomy enrichment + MERFISH cortical depth. A) GWAS enrichment per SST subtype in the 503-type RBH combined context — 17/18 SST types are FDR-significant, 10 are Bonferroni-significant. Bars colored by cortical layer: red = Upper (depth < 0.35), orange =

Middle, blue = Deep. B) Cortical depth vs GWAS enrichment. Upper-layer SST types carry stronger enrichment (Spearman $r = -0.50$, $p = 0.034$). C) Summary statistics.

SST enrichment by cortical layer (RBH combined)

Layer	Types	Mean $-\log_{10}(p)$	FDR-significant
Upper (depth < 0.35)	7	6.6	7/7 (100%)
Middle (0.35–0.55)	2	3.7	1/2 (50%)
Deep (≥ 0.55)	9	3.7	9/9 (100%)

Key finding: All 7 upper-layer SST types are FDR-significant with a mean $-\log_{10}(p)$ of 6.6 — nearly twice the mean of deep-layer types (3.7). The depth–enrichment correlation ($r = -0.50$, $p = 0.034$) confirms that SST subtypes positioned in superficial cortical layers carry disproportionately strong SCZ genetic signal, even when tested in the full 503-type RBH combined taxonomy.

4. Intersection with Case-Control Compositional Changes

Data

Case-control compositional changes from Endresz et al. (in prep) 7-cohort snRNA-seq meta-analysis of SCZ (crumblr model). This dataset reports beta coefficients estimating the change in cell-type proportion in SCZ cases vs. controls across ~100 cell types matching SEA-AD supertypes.

Hypothesis

If SCZ genetic risk is mediated through specific cell types, we might expect those same cell types to show altered proportions in post-mortem SCZ brains (either increased vulnerability leading to cell loss, or compensatory changes).

All Cell Types (n = 109 matched)

Comparison	Spearman r	p-value
GWAS beta vs composition beta	-0.181	5.92e-02
	Composition beta	vs GWAS $-\log_{10}(p)$

No significant correlation across all types. The cell types with the strongest GWAS enrichment (GABAergic interneurons) are not systematically the ones with the largest compositional changes across all types. This is expected: the composition data also captures non-genetic effects (medication, post-mortem interval, reactive gliosis) that dilute the genetic signal.

SST Interneurons Specifically (n = 18)

Comparison	Spearman r	p-value
Composition -log10(p) vs GWAS -log10(p)	0.701	1.20e-03
Composition beta vs GWAS beta (signed)	-0.494	3.70e-02

Strong convergence within SST interneurons. SST subtypes with the strongest GWAS enrichment are the same ones most affected in SCZ brains. The negative signed correlation ($r = -0.494$) indicates that the most genetically enriched SST types tend to be *decreased* in SCZ — consistent with genetic risk leading to vulnerability or depletion of these specific interneuron populations.

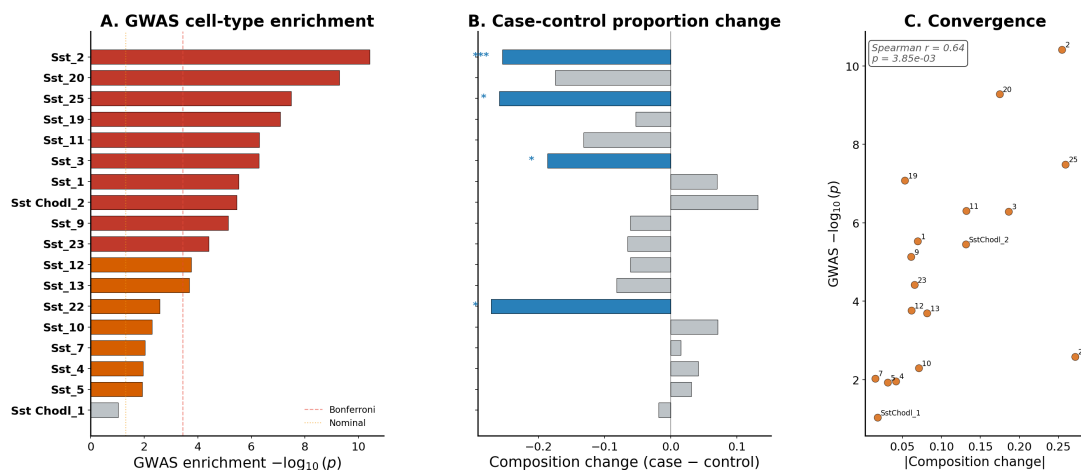


Figure 6. SST dual-hit analysis. A) GWAS enrichment per SST subtype (ranked); B) Case-control composition change — blue bars are significantly depleted in SCZ (FDR < 0.05), asterisks mark significance; C) Correlation between |composition change| and GWAS enrichment (Spearman $r = 0.64$, $p = 3.85e-3$). Subtypes like Sst_2 and Sst_25 are both highly enriched genetically AND depleted in SCZ brains.

SST Subtypes: GWAS Enrichment vs Composition

SST subtype	GWAS $-\log_{10}(p)$	GWAS sig	Comp. beta	Comp. FDR
Sst_2	10.4	***	-0.255	0.001
Sst_20	9.3	***	-0.175	0.192
Sst_25	7.5	***	-0.259	0.037
Sst_19	7.1	***	-0.053	0.640
Sst_11	6.3	***	-0.132	0.405
Sst_3	6.3	***	-0.186	0.046
Sst_1	5.5	***	+0.070	0.570
Sst Chodl_2	5.5	***	+0.131	0.390
Sst_9	5.1	***	-0.061	0.570

SST subtype	GWAS $-\log_{10}(p)$	GWAS sig	Comp. beta	Comp. FDR
Sst_23	4.4	***	-0.065	0.652

Sst_2 is the standout: the 3rd most enriched SEA-AD type for SCZ GWAS signal (FDR = 2.88e-05) AND the most significantly depleted SST subtype in SCZ post-mortem tissue (FDR = 0.001, beta = -0.255). **Sst_25** and **Sst_3** show a similar dual-hit pattern (both GWAS-enriched and compositionally depleted at FDR < 0.05).

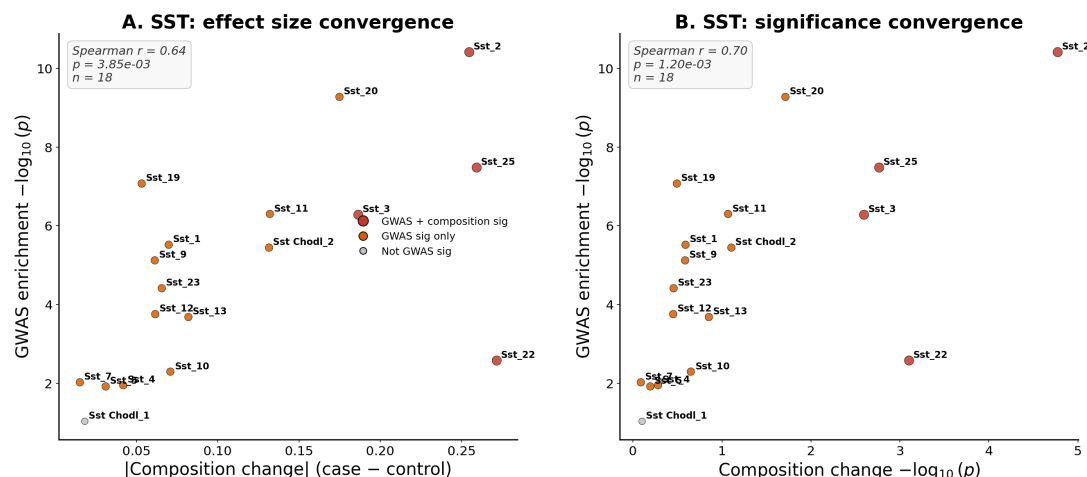


Figure 7. Correlation between GWAS enrichment and composition change within SST interneurons. A) Effect size convergence: |composition change| vs GWAS $-\log_{10}(p)$, $r = 0.64$, $p = 3.85e-3$. Dark red points are dual-hit types (both GWAS and composition FDR-significant). B) Significance convergence: composition $-\log_{10}(p)$ vs GWAS $-\log_{10}(p)$, $r = 0.70$, $p = 1.20e-3$. Sst_2 is the extreme dual-hit in both panels.

5. Interpretation

Why GABAergic interneurons?

The enrichment of SCZ genetic risk in GABAergic interneurons — especially SST and Pvalb subtypes — is consistent with the **GABAergic hypothesis of schizophrenia**: that deficits in cortical inhibition, particularly from SST+ and PV+ interneurons, produce the disinhibition of pyramidal cell circuits that underlies psychotic symptoms.

Why specific SST subtypes?

Not all SST subtypes are equally enriched. Upper-layer SST types (those positioned in cortical layers 1-3) tend to carry stronger enrichment than deep-layer SST types. The conditionally independent types (Sst_2, Sst_20, Sst_1) each carry non-redundant genetic signal, suggesting distinct biological pathways converge on different SST populations.

The case-control convergence

The convergence of GWAS enrichment and compositional depletion on the same SST subtypes suggests a causal chain:

SCZ risk variants

- > alter expression of genes specific to SST subtypes
- > impair development/maintenance of those subtypes
- > reduced SST subtype proportions in SCZ brains
- > cortical circuit disinhibition
- > psychotic symptoms

This is speculative but is the simplest model consistent with both the genetic (GWAS enrichment) and post-mortem (composition depletion) data pointing to the same cell types.

Caveats

1. The composition changes are from post-mortem tissue and may reflect medication effects or disease progression, not purely genetic etiology
2. The correlation within SST ($n = 18$) is significant but based on a modest sample size — it should be validated in independent cohorts
3. The absence of a pan-cell-type correlation ($r = -0.10$) means this is a subclass-specific phenomenon, not a general rule

6. Figure Summary

Figure	File	Description
1	results/figures/manuscript/fig1_seaad_manhattan.png	SEA-AD Manhattan scatter (137 types, grouped by subclass)
2	results/figures/conditional_analysis_results.png	Conditional analysis: marginal vs conditional significance
3	results/figures/gene_drivers/gene_driver_scatter_Sst_2.png	Gene driver scatter for Sst_2 (exemplar)
4	results/figures/rbh_combined_taxonomy_top_enrichments.png	Top 30 enrichments: SEA-AD vs novel Siletti
5	results/figures/manuscript/fig4_sst_rbh_depth.png	SST deep dive: RBH combined enrichment +

Figure	File	Description
		MERFISH depth (3-panel)
6	results/figures/manuscript/ fig2_sst_dual_hit.png	SST dual-hit: GWAS + composition side-by-side
7	results/figures/manuscript/ fig3_sst_composition_correlation.png	SST composition correlation (2- panel)

7. Key Output Files

File	Description
results/tables/ seaad_magma_scz_enrichment_all_supertypes.csv	SEA-AD enrichment (137 types)
results/tables/rbh_combined_enrichment.csv	RBH combined enrichment (503 types)
results/tables/ independent_supertypes_conditional.csv	Conditional analysis (18 types)
results/tables/ gwas_vs_casecontrol_composition.csv	GWAS vs composition merged data

8. Scripts

Script	Description
scripts/ 01_compute_specificity.py	Compute cell-type specificity from SEA- AD snRNA-seq
scripts/ 02_magma_enrichment.py	MAGMA-style gene property analysis
scripts/ 03_conditional_analysis.py	Forward selection for independent types
scripts/ 10_combined_taxonomy.py	Build RBH combined taxonomy and enrichment
scripts/ 11_gene_driver_scatter.py	Gene driver scatter plots
scripts/ 13_gwas_vs_composition.py	GWAS vs case-control composition analysis
	Generate figures for this document

Script	Description
scripts/ generate_markdown_figures.py	